An Introduction To Type-E Chopper

1.1 General

In case of four-quadrant chopper, the load current as well as load voltage are either positive or negative which provide the four quadrants of operation (shown in Fig. 1.1). This chopper is also known as "Type-E chopper" or "Full-Bridge DC-DC Converter". It consists of four (power) semiconductor switches along with an anti-parallel diode across the each switch. The numbering of choppers corresponding to their respective quadrant of operation. E.g. for first-quadrant operation, only switch-CH1 is modulating and for second-quadrant operation, only switch-CH2 is modulating and so on. It should be noted that for operation in the third and fourth quadrants, the direction of the load emf E must be reversed internally. There are three distinct application of the full-bridge switch-mode converter as follows:

- DC-Motor Drive.
- DC-to-AC (sine-wave) conversion in single phase uninterrupted ac power supplies.
- DC-to-AC (high intermediate frequency) conversion in switch-mode transformer-isolated dc power supply.

Here the load is a DC motor (i.e. it has resistance, inductance and a dc voltage source). We assume that the DC voltage source is ideal. Two switches of same leg must not on simultaneously, otherwise short circuit across the source will be occurred.

1.2 First Quadrant:

For first-quadrant operation (in Fig. 1.1), CH4 is kept on continuously, CH3 is kept off continuously and CH1 is modulated. With CH1 and CH4 both are on condition, load voltage $v_0 = V_s$(source voltage) and load current(i_0) begins to flow from node-a to node-b i.e. positive. Again when CH1 is turn off, positive load current(i_0) freewheels through CH4, D2, Load and CH4 path. Hence, the output quantities (i.e. V_0 and I_0) can be controlled in first quadrant and they are both positive which implies that the power flows from source to load i.e. motor works as "forward motoring" operation. It is also observed that this converter behaves as a "step-down" chopper in this quadrant.

1.3 Second Quadrant:

Here switch CH2 is only modulated and other switches (i.e. CH1, CH3 and Ch4) are kept off continuously (in Fig. 1.1). When CH2 is on state, negative(or reverse) current flows through load, CH2, D4 and load and load inductor L stores energy during this interval.When CH2 is become off, current is fed back to source from load through D1 and D4 , hence, stored energy of inductor is reduced. It is noticed that here load voltage (i.e. $v_0 = E + L\frac{di_0}{dt} + Ri_0$) is more than source voltage (V_s). As the load voltage is always positive whereas load current is negative, which gives operation in second operation and power is fed back to source from the load. The motor works as in "forward regenerative breaking" mode. The converters acts like a step-up chopper.

1.4 Third Quadrant:

Here CH1 is kept off continuously, CH2 is kept on continuously and CH3 is operated (in Fig. 1.1). The polarity of the load emf E must be reversed internally to have a successful operation in this quadrant. When CH3 is on, the load gets connected to the source, hence, load voltage and load current both are negative. Again when CH3 is turned off, the negative current freewheels through CH2, D4, load and CH4 path leading to third quadrant operation. The power flows from source to load and the aforesaid quadrant the chopper behaves like a step-down converter. The motor acts in "reverse motoring" operation.

1.5 Fourth Quadrant:

In this quadrant only CH4 is modulated and others switches are kept off stage continuously and the polarity of the load emf E must be reversed internally as stated in the previous quadrant of operation (in Fig. 1.1). When CH4 is on, positive load current flows through the path CH4, D2, load and CH4 and the load inductor stores energy during this interval. Again CH4 is turned off, load current is fed back to source through D2 and D3. Hence, the load current is always positive (i.e node-a to node-b) but load voltage is negative which leads to fourth quadrant of operation. It acts like a step-up chopper in this quadrant. The motor is in "reverse regenerating breaking" mode of operation.

Hence, the operations of the type-E chopper can be summarised as shown in Fig: 1.2.

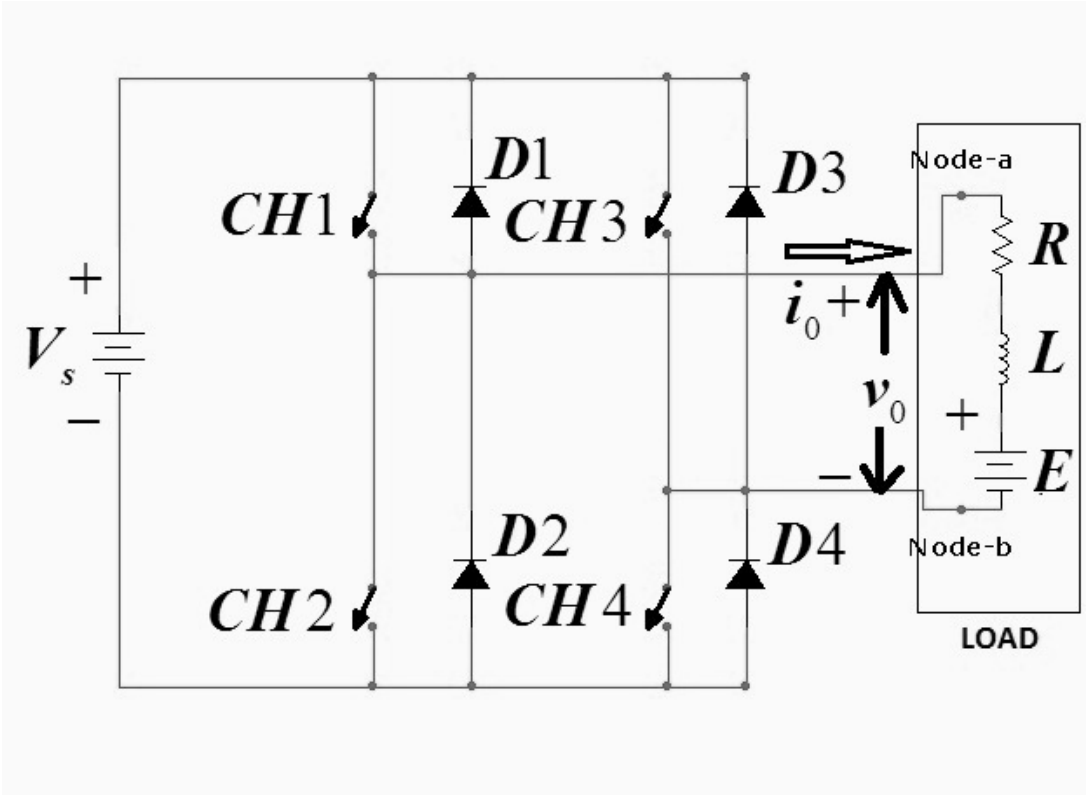


Figure 1.1 Circuit Diagram of a Type-E Chopper

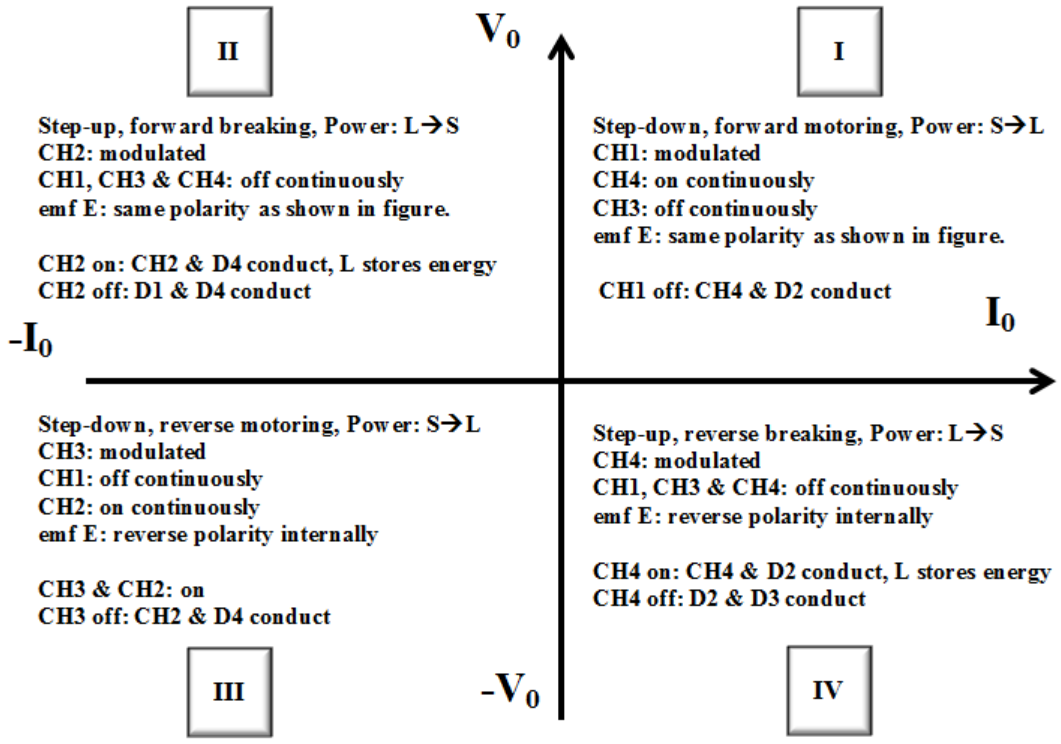


Figure 1.2 Conduction and polarity of devices of a Type-E Chopper